California Polytechnic State University, San Luis Obispo

Kennedy Library Occupancy Counter

Kylie O'Donnell

Senior Project

Dr. Michael Haungs

June 2024 - December 2024

Introduction

As of the 2023-2024 school year, the Kennedy Library has been undergoing renovation which has made finding comfortable and quiet on-campus study spaces difficult. Currently, there are temporary study locations that allow students to find areas around campus with their associated occupancy counters. However, the current spaces that are being tracked are inconsistent and it isn’t in an accessible part of the Kennedy Library website making it hard for students to find. The occupancy counters would be useful in midterm and final season, as it would allow students to have an idea of which place might be best to study. With this in mind, it will be beneficial to have occupancy counters in the remodeled Kennedy Library to show students which floors are more occupied than others and also show which individual study rooms are full or empty. As a student who has walked around multiple floors of the Kennedy Library and found no open space, occupancy counters would have saved me time if I had known which floors had available spaces and would have helped plan out my study times.

I believe that implementing an occupancy counter in Kennedy Library would be beneficial to students to help plan their study days and give them insight into what the library looks like at certain times. Furthermore, implementing the occupancy counter on the Kennedy Library homepage would help drive more traffic to the website while also allowing an easy way for students to check the occupancy of floors and rooms.

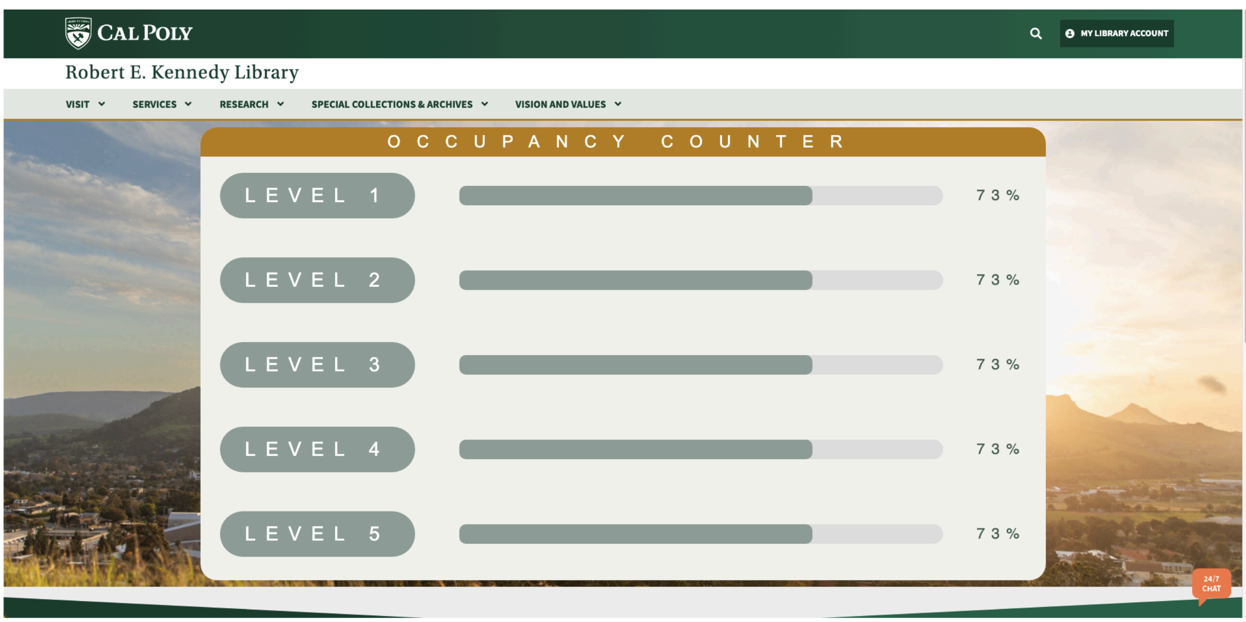
Currently, our school uses Occuspace which is an anonymous sensing device to deliver accurate and real time occupancy data. Currently there are other occupancy counters such as physical ones (turnstiles), and other occupancy counters that use cameras. However, Occuspace is the best device because it comes with its own dashboard, API, and keeps students and other members of the community anonymous.

My work specifically involves creating an occupancy counter for the new Kennedy Library so students can easily view the occupancy of the library at any given time. This would allow students to plan their study times without having to go to the library in-person. Too many times I have heard from my peers that they go to the library to find a place to study, for it to be too crowded and not be able to find a place to study (which ultimately wastes time that could be used for working on assignments/studying). The occupancy counter that I created works best when shown on the homepage (or has a link to on the homepage) of the Kennedy Library website. The occupancy counter shows users an overall view of occupancy levels for all five floors. When the user clicks to each floor, they will be directed to a more in-depth view of the occupancy count and will be able to view what parts of that floor are crowded, and what floors are free. Furthermore, from the in-depth view of each floor, students will be able to view and book available study rooms.

After the occupancy counter was created, user testing was done to see what users liked and what users were confused about. Many users were originally confused about the colors to depict occupancy, so clarification was provided on the webpage. Some other confusions regarding wording were also brought up, but the occupancy counter was iterated based on the feedback. Overall, user testing showed that many users (student and faculty members) find the occupancy counter extremely useful. One user when asked if they would find it useful to find available space in the library said "absolutely because it would help me decide where to go. I could know immediately if a place was completely full".

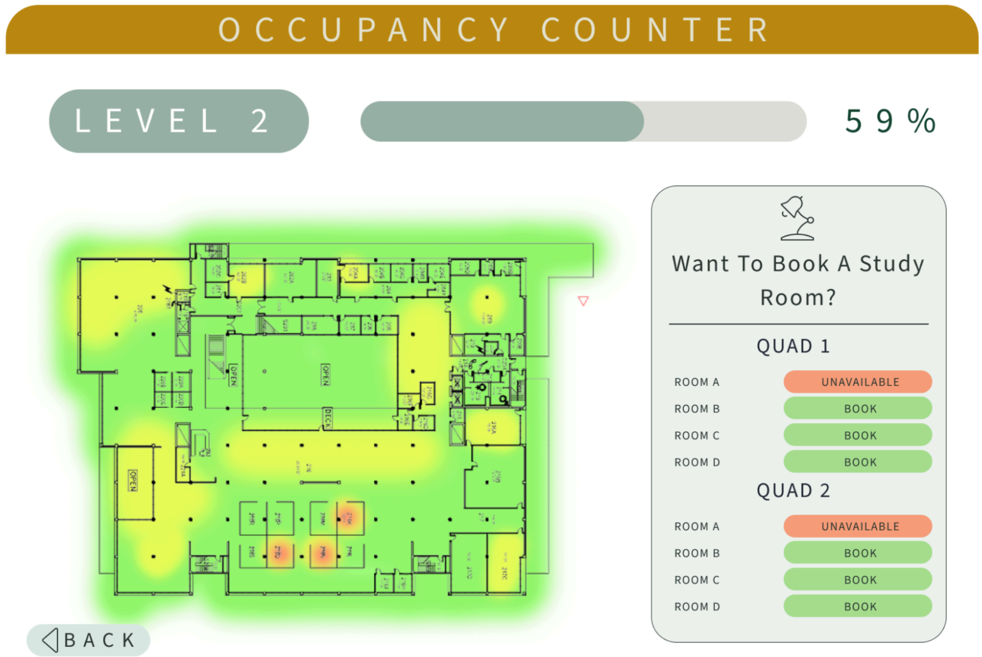
Application

To access the occupancy counter, students will navigate to the Kennedy Library homepage and click to view the occupancy counter. On an overview level, the occupancy counter will show students level capacities of the library at the current time. This will allow students to quickly determine if they want to visit the library or not, and gives them a sense of the crowd at the given time.



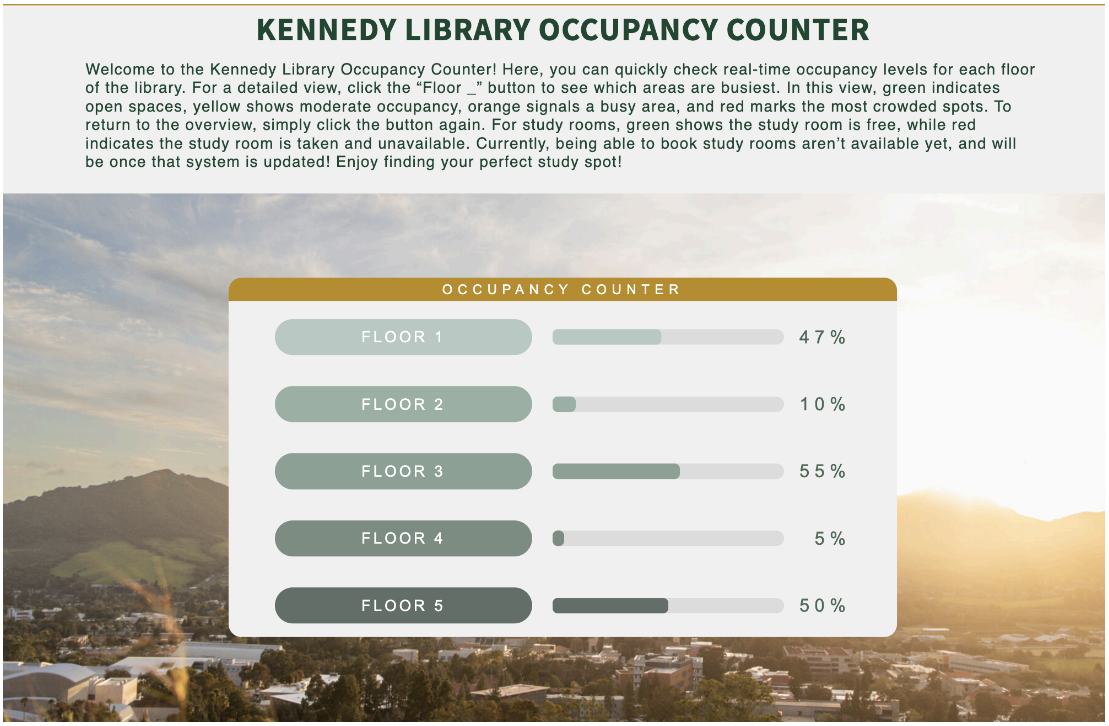
Initial Prototype Design

If a student wants to know more information about the occupancy counter, the student will just click one of the levels to be brought to a page that shows the breakdown of the floor. The breakdown of the floor will detail any study rooms that are available or taken, and will allow for students to book the available ones. Furthermore, the breakdown of the floor will show which parts of the floor are more crowded than others to help students find areas that might have open seating.

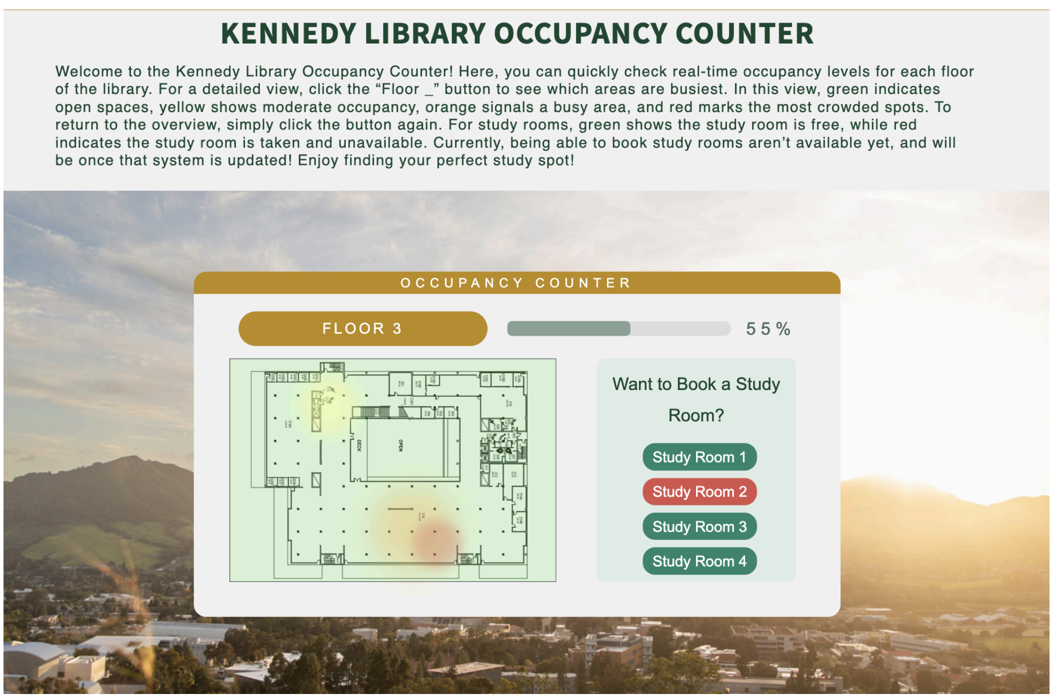


Initial Prototype Design

After some user feedback, the initial prototype was iterated on and the final design is shown below. Instead of a back button, users click the "level __" button again to go back to the overview. Furthermore, each study room button just has a green or red button to show if it's available to book or unavailable. Lastly, the full occupancy counter page does have a brief description to clear up any confusion regarding how to navigate the occupancy counter, and what each color means on the floor map. More about changes regarding the initial prototype design and the finished occupancy counter design is explained in Analysis / Verification.



Finished Occupancy Counter Design. This shows the overview of each floor occupancy.



Finished Occupancy Counter Design. When clicking into a floor level, it opens into a more detailed view of the floor, showing which parts of the floor are more crowded than others. It also allows users to book study rooms.

Background

Researching other colleges and their use of occupancy counters, the main technology that is used is with a company named Occuspace (https://web.occuspace.io/). Occuspace uses sensors that detect Bluetooth Low Energy and Wifi signals within an area. Unlike other sensors that use different types of camera systems, the good thing about this sensor technology is that it doesn't collect any personally identifiable information. This allows for student's and other guest's information to remain private and secure. According to it.wisc.edu, one sensor can cover up to 5,000 square feet. In addition to the actual physical Occuspace Sensors, they come with their API to integrate it into custom web pages.

Using the Occuspace API (https://occuspace.io/api) to gather real-time occupancy estimates for each location, the API will be embedded into an Occupancy Counter window within the Kennedy Library homepage. The Occuspace API returns information about each location in a JSON format and tells real-time and historical count estimates.The API gives information regarding accessible locations, real-time data on the most recent estimated count for the requested location, daily occupancy, hourly occupancy, daily and hourly visitors, and daily dwell time. If enough Occuspace devices are planted on each level, getting a good sense of what specific areas are more crowded than others can allow for students to view what parts of each level are more crowded than others. The historical count estimates are a useful tool to help provide estimated guesses on how busy the library might be during a certain time based on pattern. This will allow students and other community members to easily access and find space data in real time. Furthermore, designing a mockup of the Occupancy Counter window will ensure the occupancy counter will be user-friendly and easily navigable to users.

The occupancy counter will be drafted using Figma, which allows for designs to become functional prototypes. Figma is a web-based design application that allows designers to create interactive graphics and wireframes to show and test designs of websites, mobile applications, and more. After drafting the graphics of what the occupancy counter would look like, Figma allows each element in the graphic to become a component that can be interacted with. Connecting these components, also known as wireframes, will allow me to show the library my current design and will allow for collaboration to make the occupancy a seamless experience for the user. After finishing the Figma design, the occupancy counter window will first be implemented in WordPress, the current space where the Kennedy Library pages would be hosted.

WordPress is a web content management system that allows for the customization and creation of a webspace. WordPress has easy components that allow novice web developers to easily drag and drop components without any coding. WordPress also allows for the implementation of custom components by implementing them in HTML. Since I will be using a design created in

Figma, implementing the Occupancy counter will be combining both HTML for style choices as well as the Occuspace API. Despite WordPress's easy accessibility to novice web developers, WordPress doesn't allow for much customization and can be more of an obstacle to developers who are used to full customization. Despite this, WordPress does allow the use of API's so getting data from the Occuspace API and showing them on the Kennedy homepage should be seamless.

Lastly, user testing will be conducted to see feedback from both faculty members and students. This feedback will be used to enhance the design of the occupancy counter and make sure that it's accessible and easy for users to get information. User testing will be conducted by hosting a live version of the prototype, and asking the tester a series of questions after they navigate around the counter. User testing will consist of four sections: General Usability, Visual Design, Functionality, and Overall. This is to get an in-depth depiction of how users view the Occupancy Counter.

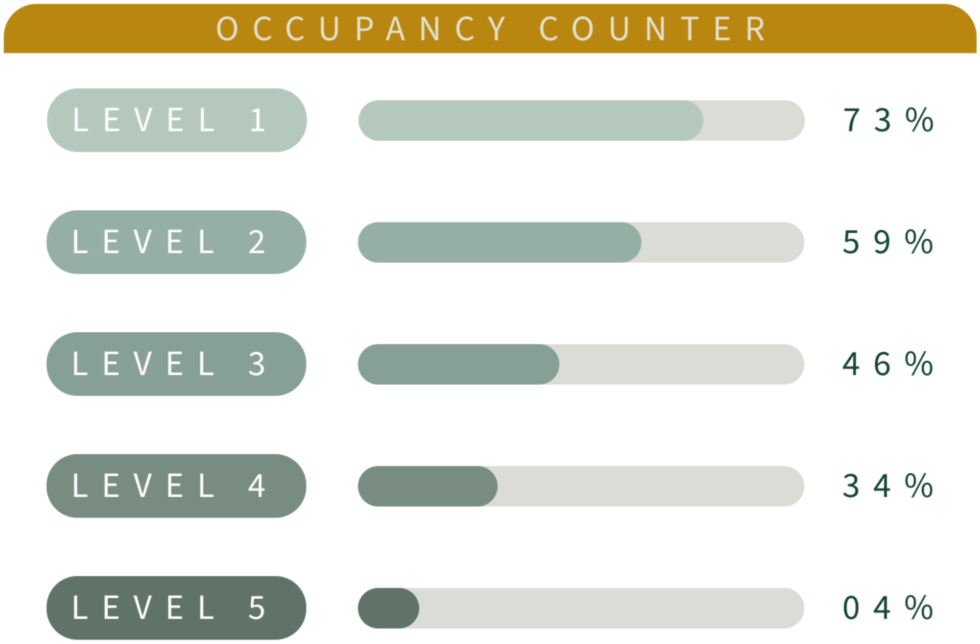
Design

The main goal of my project is to implement an occupancy counter that is user-friendly within the Kennedy Library website. With the current renovations of Kennedy Library, the use of the website is limited and there is a big push to get students to interact with the library website, especially when it opens up. There are current occupancy counters that are set up in temporary study space locations, but it doesn't give much insight into all the locations and isn't user-friendly.

TEMPORARY STUDY SPACE LOCATIONS			
Building (building #)	**Rooms**	**Current Availability**	**Schedule**
1901 Marketplace (19)			For current hours, visit Campus Dining.
Cotchett Education (02)	203 & 204		M-Sun 8:00 a.m. - 10:00 p.m.
Engineering West (21)	237 & 238		M-Sun 8:00 a.m. - 10:00 p.m.
Fisher Science (33)	First floor lounge		M-F 7:00 a.m. - 10:00 p.m.
Mathematics and Science (38)	227		M-Sun 8:00 a.m. - 10:00 p.m.
Temporary Structure (H11)	Parking Lot H11		Closed for summer break: July 5 - September 20.
University Union (65) Hub24 sponsored by ASI	2nd Floor - Chandler Lounge	**0% Full**	Visit ASI Facility hours webpage for special hours and ASI space activity webpage for more updates.
	2nd Floor - Open Space	**0% Full**	
	2nd Floor - San Luis Lounge Computer Lab	**0% Full**	

Current Occupancy Counter for Temporary Study Space Locations

This is the current look of the occupancy counter that is shown on the Cal Poly Website. It's hard to find because it's not hosted on the library homepage, but is hosted on the Administration & Finance Cal Poly page. Using the Occuspace API, implementing real-time occupancy data onto the Kennedy Library homepage will significantly help the student body locate and figure out how busy the library is. The occupancy counter will show the levels of the floor and overall occupancy. From there, the user will be able to click on each level which will bring up a more in-depth occupancy overview. This will include booking study spaces on the selected floor and which area of the floor is more crowded/less crowded.



Mockup of Occupancy Counter Window

Implementation

This project is a web-based occupancy counter that displays the occupancy levels and availability of study rooms in the Kennedy Library. The implementation of this occupancy counter uses HTML, CSS, and JavaScript to create an intuitive, interactive, and user-friendly interface.

The key components of this project are as follows:

HTML: The HTML structure serves as the backbone and foundation of this application. The main elements of the UI include background, top-bar, level-container, and info-container. These elements together are used to organize and display the content dynamically generated by the JavaScript code.

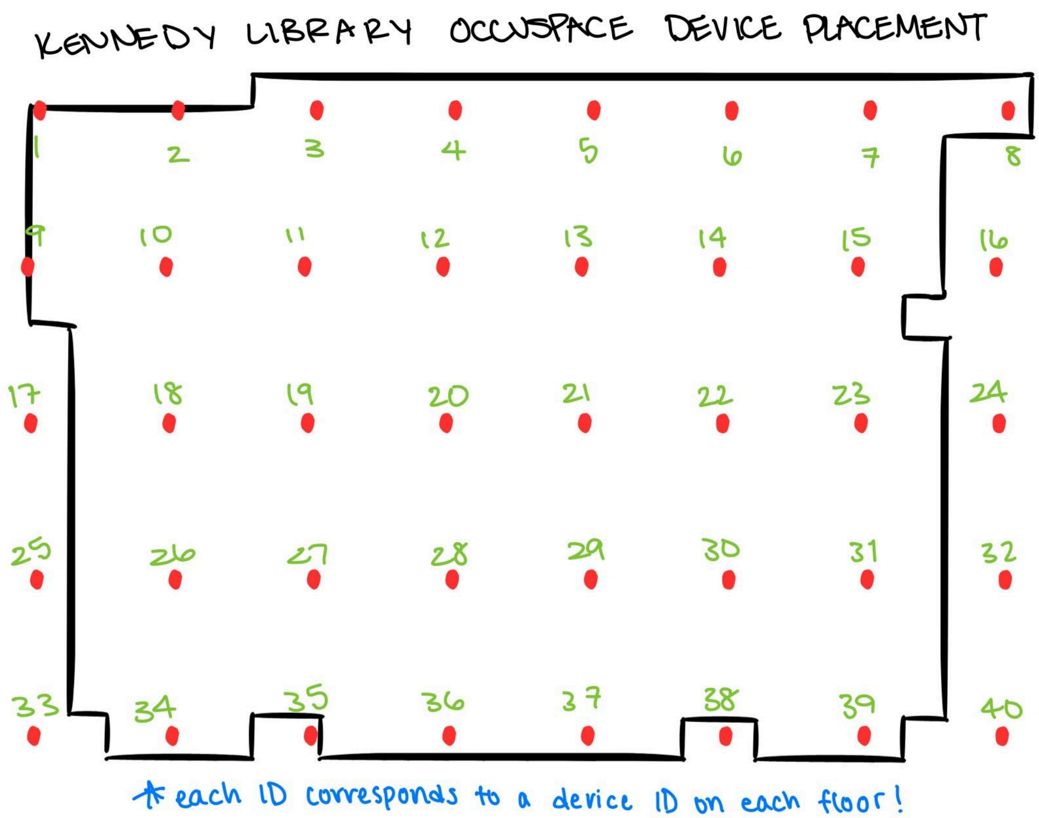
CSS Styling: CSS was used to style the various elements within the page to provide a cohesive and visually appealing design. Furthermore, flexbox was used to align and center items to make the layout responsive to different screen sizes.

Javascript: Data is represented using JavaScript objects and arrays. The levels array containers the objects representing important data from each floor including level and percentage. On the first couple iterations of the prototype, there was only one set of data. This was to make sure buttons and general use of the prototype was working.

In the code snippet below, the study1 array contains objects representing the data from each study room and whether they are available or not. The study1 array only applies to the study rooms on level one, and there are more study rooms that reflect the other levels. Furthermore, there is floor data that correspond to different spots on each level. Each of these spots have an associated "id" number that is the same from floor to floor. These id's have a "crowdLevel" number that determines the occupancy in that certain area and corresponds with a heat map color. This allows for students to view the occupancy data in a more in-depth view. The best way to have the most accurate data and information at each spot on the floor is to have around 40 Occuspace devices. When creating the Occupancy Counter, I initially started with 1 dataset, making my prototype static. However, due to not having access to Occuspace devices, mocking a rotational dataset was imperative to show users how this counter would work with changing occupancy numbers. To do this, I implemented three datasets to show how the prototype would realistically work. The dataset would change every ten seconds and re-renders the web page to ensure that all the data was updating live. In the code snippet below, it shows "dataSetC" which is one of the three datasets that is rotated through the occupancy counter.

```
name: 'dataSetC',
                data: {
                    floors: [
                        { floor: 1, percentage: 47, color: '#B6CCC2', hoverColor: '#BD8B13'},
                        { floor: 2, percentage: 75, color: '#98B3A7', hoverColor: '#BD8B13'},
                        { floor: 3, percentage: 55, color: '#8BA499', hoverColor: '#BD8B13'},
                        { floor: 4, percentage: 30, color: '#798F85', hoverColor: '#BD8B13'},
                        { floor: 5, percentage: 50, color: '#61736B', hoverColor: '#BD8B13'}
                    ],

                    study1: [
                        { room: 1, availability: true }, { room: 2, availability: false },
                        { room: 3, availability: false }, { room: 4, availability: true }
                    ],

                    study2: [
                        { room: 1, availability: false }, { room: 2, availability: false },
                        { room: 3, availability: false }, { room: 4, availability: true }
                    ],

                    study3: [
                        { room: 1, availability: true }, { room: 2, availability: false },
                        { room: 3, availability: true }, { room: 4, availability: true }
                    ],

                    study4: [
                        { room: 1, availability: true }, { room: 2, availability: true },
```

```
            { room: 3, availability: true }, { room: 4, availability: true }
        ],

        study5: [
            { room: 1, availability: false }, { room: 2, availability: true },
            { room: 3, availability: true }, { room: 4, availability: false }
        ],

        floorData: {
            "floor1": [
                { "id": 1, "x": -20, "y": 100, "crowdfloor": 1 },
                { "id": 21, "x": 200, "y": 240, "crowdfloor": 2 },
                { "id": 31, "x": 450, "y": 400, "crowdfloor": 3 },
                { "id": 34, "x": 100, "y": 460, "crowdfloor": 2 },

            ],
            "floor2": [
                { "id": 10, "x": 45, "y": 160, "crowdfloor": 4 },
                { "id": 11, "x": 330, "y": 370, "crowdfloor": 3 },
                { "id": 11, "x": 200, "y": 360, "crowdfloor": 2 },
            ],
            "floor3": [
                { "id": 11, "x": 150, "y": 160, "crowdfloor": 2 },
                { "id": 29, "x": 320, "y": 410, "crowdfloor": 3 },
                { "id": 30, "x": 370, "y": 440, "crowdfloor": 4 }
            ],
            "floor4": [
                { "id": 27, "x": 130, "y": 420, "crowdfloor": 3 }
            ],
            "floor5": [
                { "id": 19, "x": 210, "y": 320, "crowdfloor": 2 },
                { "id": 27, "x": 320, "y": 430, "crowdfloor": 3 },
                { "id": 28, "x": 200, "y": 400, "crowdfloor": 4 }
            ],
        }
    }
}
```



Blueprint Depiction of Floor ID Numbers and Occuspace Devices

In addition to this, Javascript is also used to create and manipulate DOM elements based on the given data. Buttons are added to handle user interactions such as clicking on a level button to display more information. Lastly, there are `onmouseover` and `onmouseout` events to change the background color of buttons when hovered over, to provide visual feedback to the user.

Lastly, I have complex interactions that are coded using Javascript. When a level button is clicked, the page hides the level-container and displays the info-container to provide detailed information about the selected level. The DOM elements are cloned so I can still display the selected level's information (occupancy percentage and study room availability).

The general logic for how the occupancy works is seen in this pseudocode:

```
FOR each level in levels
    CREATE level button
    SET button background color
    ADD hover effect to change background color
    CREATE slider container and fill based on occupancy percentage
    APPEND button and slider to level container

    ON button click:
        HIDE level container
```

```
        SHOW info container
        CLONE button and slider to details container
        LOAD and display floor image
        CREATE and position roundedRectangle next to floorImage
        FOR each study room in study1
            CREATE study room button
            SET availability status (available/unavailable)
            APPEND to roundedRectangle
        END FOR
        POSITION roundedRectangle based on floorImage
END FOR
```

Analysis / Verification

To determine if this project was successful, there are several aspects that need to be considered: scalability, efficiency, correctness, and user experience.

Scalability can be tested by adding more levels, rooms or data points to the system. This will be mostly added with adding more sensors to give a more accurate reading of each level on the library floor.

Efficiency can be tested and measured by analyzing the time between when the sensors load and when it updates live to the website. The occupancy counter should be updated regularly to give real-time updates.

Correctness is determined by verifying that the application accurately reflects the current occupancy data and room availability. Automated tests or manual verification can be used to ensure the data is correctly displayed.

Conducting user testing is critical to ensure the application is intuitive and meets user needs. During testing, users are asked to complete tasks, such as checking the occupancy of a specific floor or finding an available study room. Observing how easily they can complete these tasks will provide valuable insights into the application's usability. The feedback questionnaire can be viewed at this link.

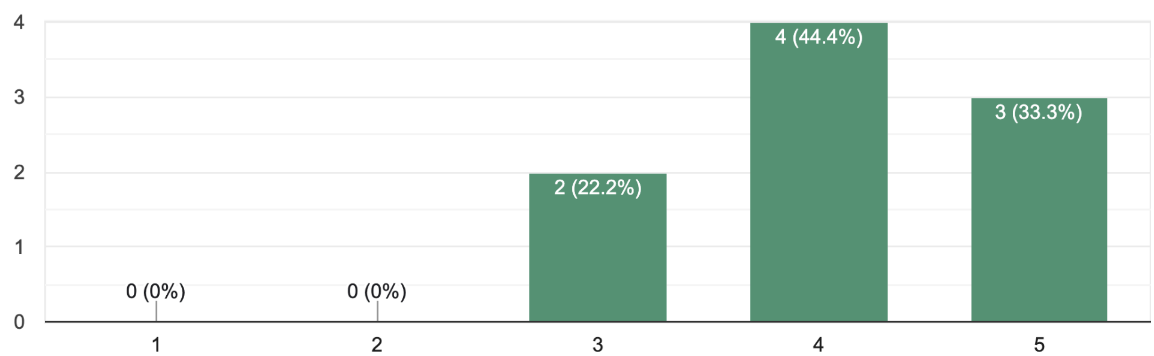
User testing was conducted during week 3 of Senior Project II and continued through week 9. The prototype was hosted on WordPress under a custom HTML button and feedback was given through a series of questions the user would go through with the prototype. The feedback form consisted of 4 sections: general usability, visual design, functionality. and overall thoughts. In each section, I had users rate the usability of the prototype, if they encountered any issues, and if there was anything that was unclear/confusing in the prototype.

Initial responses noted that the interface was intuitive and easy to use but the heat maps to convey occupancy in certain areas were unclear to some. Many users noticed they were confused on the coloring with the heatmaps and were wondering if the connectedness of the heatmaps also had to do with the study room buttons. To fix this confusion, I added clarification at the top of the occupancy counter, to let users know what the colors on the floor plan meant as well as what the colors on booking study rooms meant. Furthermore, there were some notes regarding the UI and deleting the back button, making the UI a tad bit more standard (like Apple navigation menus), and some slight typography re-work. After some iterations, the final design includes rotational data, heat maps, and booking a study room (this is currently not functional, but would ideally work when the library's system is back in place!).

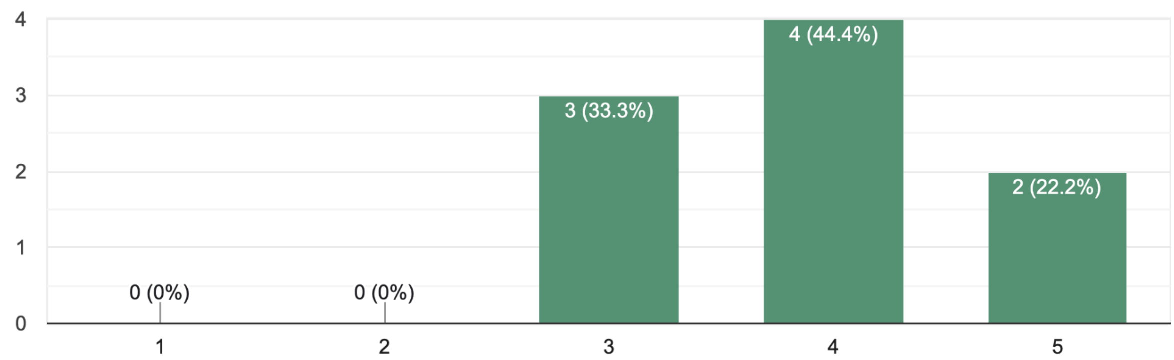
Overall, users gave my design an average of 4.11 stars out of 5. The average rating includes understanding the purpose of the occupancy counter, how easy it was to locate information they were looking for, overall layout and visual appeal, and usability with navigating the window. Below depicts the ratings that users rated different aspects of the occupancy counter .

On a scale of 1 to 5, how easy was it to understand the purpose of the occupancy counter? (5 being the easiest)

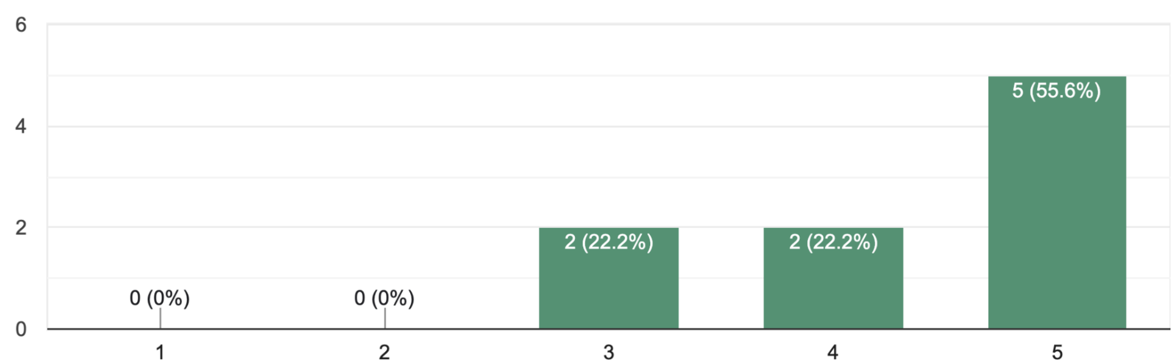
9 responses

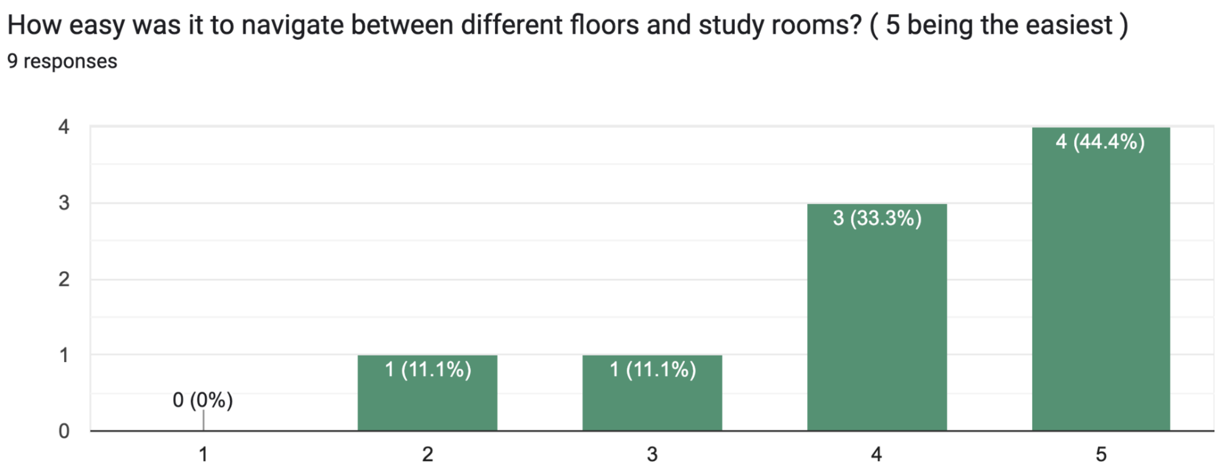


How would you rate the overall layout and visual appeal of the occupancy counter? (5 being omg this is amazing!)
9 responses



How easy was it to locate the information you were looking for (e.g., available rooms, occupancy levels)? (5 being the easiest)
9 responses





Short Answer Questions:

Did you find the interface intuitive to use? Why or why not?	Yes. It let me see every floor and give a general idea of how full each of the floors are and where a possible place to be is open. Yes it's nice that you can click each level and get the separate map for each. I don't quite understand what the yellow and red spots are for though. It is also quite hard to read the map because the words are small and blurry. I like that it is very clear which rooms are available or taken from the list on the side. At first I didn't know what to click on, but once I clicked on a level, it became clear what I was looking at. It might be good to add some sort of description so it's clearer to a first-time user. You could use a cookie to display it until they dismiss the message. Yes it was intuitive because the layout is simple and the buttons are self-explanatory Yes, I did find the interface intuitive to use because the layout is very organized and easy to understand. The gradations of color for each floor are confusing since they don't align with percentage of occupancy. Overall, yes! I was just confused on how to return to the overall list after clicking on a specific level. Yes easy to use but needs a few tweaks to make it clearer on how to use the booking tool. The initial view is intuitive enough, but when I click on a floor I don't understand how the percentage for the floor is connected to the floor

	plan map or the study room buttons (actually, I'm not sure those are buttons?) The floorplan map in general doesn't seem intuitive to me, I'm not sure what the colors mean.
Were there any moments where you felt confused or unsure of how to proceed? If yes, please describe.	One thing that confused me is what is the percentage based off of. Is it based of the total amount of people allowed on the floor during an emergency or is it available seating I was confused about the orange/yellow blurs on the map. I thought that they were related to the study rooms but I think they might just be busy areas of the library? I do not attend this school so maybe it would be more clear if I had been to this building before Only on initial page load - see first response for more no I was a bit confused about the coloring on the maps like for the levels, but I think it is referring to which space in the library is being occupied. Why did you use "level" instead of floor? First Floor, Second Floor, all seem more intuitive than the word level. After clicking on Level 1, I was unsure of how to return to the overview page. Are the red/green colors next to each other accessible? I figured it out fairly quickly, that you have to click the level again to go back to showing all the levels and to book the room you would click the green button. It might benefit from having directions written out, too. What do the colors on the floorplan maps mean? How is the floorplan map connected to the study room buttons? Are the study room buttons even buttons that should be clicked? What do the colors for the study rooms mean?
Did the colors and icons used to represent occupancy levels (e.g., dots for crowded spaces) make sense to you?	Yes red means the area would be occupied like how maps shows red to be have traffic not sure what icons are being referred to in this questions but I already expressed some initial confusion in one of the previous questions Yes the heatmap is very intuitive for showing busy-ness Yes I think the colors are awesome and helpful! For the most part, after looking at the colors and icons a bit more, it does make sense to me, but was confused a bit in the beginning. It isn't intuitive that you click on the LEVEL to get more data. I didn't know that until I got to this question. I can't really tell the occupancy using the maps. Yes; a key for color indication would be useful.

	Yes! I'm not sure what the colors on the floor plan map are supposed to mean.
Was the size of the text and buttons appropriate for easy interaction? If not, what would you change?	Yes The only thing that felt slightly out of place was the "back to levels button. I appreciate the wording for making it clear how to get back to the "home" page and I feel like the location logically makes sense but it feels squished between the header and the main program. I think the typography could use some work. The spacing between the letters is a little distracting for me, for example. Yes Yes You need to show somehow that things are a button. Some instructions would help. Red=busy, very pale, hard to see yellow is somewhat busy, etc. Yes Yes, it looks great. They seemed appropriate
Did you encounter any issues while interacting with specific features (e.g., viewing room availability or switching floors)? If yes, please briefly explain.	Yes study room buttons dont work looks great to me Going back to the main menu to view a different floor is a bit jarring - I would adjust the size of the back button, or make the UI more like a standard card (see apple navigation menus for inspiration, they are very clear about intended action) no No When I got to a specific LEVEL I didn't know how to get back to the original full page of levels. Initially, I was confused on how to switch floors. But once I clicked around, it made sense. Maybe a back arrow would be helpful on individual level pages. Explained earlier but I did figure out pretty quickly that you press the level button again to go back to all levels. I wonder if that might confuse others?

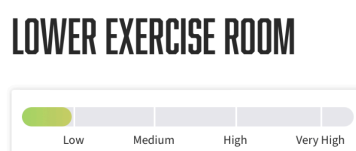
	As mentioned before, I'm not sure if clicking on the study room buttons is meant to cause something to happen on the page.
Do you feel the occupancy counter would help you find available spaces in a real setting? Why or why not?	Yes Yes I like this tool. I think it's nice to know which floors are the least busy and it seems like an easy way to book a study room If implemented properly, it would help - I am not sure how you would implement though. One option is to have users check themselves in and where they are going, but this would be unreliable. Another option is thermal and movement detection cameras scattered throughout the building for automatic detection, but this would be expensive and possibly unmaintainable. Useful? Yes, very. I would like to see it done if technologically possible. absolutely because it would help me decide where to go. I could know immediately if a place was completely full Yes, because there are times when I walk into the library and waste time looking for a free table and this is a good resource to use, to check which places are crowded. But, I am interested in knowing how the data is collected, like is this collected with past information or like it is live updates of the occupancy at the library. Perhaps helpful if the colors were brighter to show occupancy. Yes, the "heat" mapping on the level pages made sense. I would like the floor plans to be less detailed though. Like I don't think it needs to show the electrical closets, etc. Yes, it shows what space is available. Yes, it could, because it gives a quick overview of the floors' occupancy levels
What did you like most about the occupancy counter?	The area marked by colors to see "traffic" The red/green indicators for available study rooms It's a nice quality-of-life feature that is missing from campus The floorplan with the heatmap and the overall % occupancy I like that the occupancy counter is quite straightforward and it is clear how to navigate. That it exists is good. Clean design The landing page where it shows the occupancy of all levels all at once.

	The initial view, which gives a good idea of how full each floor is
What aspects did you find frustrating or difficult to use?	None n/a The typography needs work, see a previous answer for more information It felt a little bit weird to have to go back to the main menu to change levels, see suggestions… An aspect that I find difficult to use would be the button for the study room, but I do understand it is in the process of development. Navigation, lack of instructions, lack of intuitive design. The green background color of the floor plans was a little confusing with the availability colors. Navigating from an individual level to all the levels. I also didn't realize at first that you would click the room button to book. May need to add written directions? See my previous answers
Do you have any suggestions for improving the interface or functionality of the occupancy counter?	It performs its function but it looks bland Just a small aesthetic suggestion but maybe consider making the "occupancy counter" header a little taller. it feels like the font is a little too close to the edges. I think the functionality is excellent because there aren't too many buttons and it's easy to pick up on how to use it. I also like how the percentages make it easy to determine the occupancy without having to click anything. good job! See a previous answer for more information I think it could be beneficial to implement a menu to go to any level from any floor instead of having to go back to the main menu. (But the interface is pretty good as is rn :)) A suggestion to improve the interface is to have a redirection page back to the occupancy counter like if I were to click another link in the website, I am not sure how to navigate back to the occupancy counter. Replace "level" with First Floor, etc. Make the colors of the occupancy brighter, have some small amount of instruction or redesign to make interaction more intuitive. Just the ones already mentioned. I think the interface is great for the most part and I provided comments about functionality in previous responses.

	Not at this time

Related Work

Current work that is similar first includes the current occupancy counters on the ASI Current Space Activity webpage. This webpage implements occupancy counters for select ASI-managed facility spaces. This includes rooms in the recreation center, in the University Union, and some lounges around campus. The task bar fills up with green and designates four types of occupancy degrees: low, medium, high, and very high. Although this is very insightful, the occupancy counter is hidden and many of my peers didn't know this existed. My work differs from this work because I will be implementing an occupancy counter that deals with the different floors within Kennedy Library while also showcasing what study rooms are available and being used. Since the Kennedy Library is much more expansive, having as much information to give to students about what is occupied and unoccupied is important to help users of the library be prepared before entering the library. My occupancy counter would also be best used if placed on the homepage of the Kennedy Library so students know that we have one.



Cal Poly ASI Facility Space Occupancy Counter

Another quick work that is very similar is library occupancy counters for libraries at other colleges. One example of this is Cal State LA's Library Occupancy counter. Its occupancy counter and table reminds me much of how our temporary library space counters work. It records the current occupancy of each library level. One thing that I will be implementing in my work would be showing what parts of that library level will be occupied and what is free. I will be giving students and staff more insight to help plan their library trip. Although the Cal State LA's Library Occupancy counter is insightful, it lacks more information about what parts of the level are busy and free. During exam time, the levels will be more full, so just showing percentage doesn't provide enough information to students.

CURRENT OCCUPANCY - LIBRARY NORTH BUILDING

Library North Levels	Max Occupancy	Current Occupancy	Percentage Available
Total Count Occupancy of all Floors	3168	53	98% Available
Library North Level 4 & Roof	5	NaN	100% Available
Library North Level 3	659	0	100% Available
Library North Level 2	659	37	94% Available
Library North Level 1	435	3	99% Available
Library North Level B	700	8	99% Available
Library North Level A	657	5	99% Available

Cal State LA Library Occupancy Counter

Future Work

Currently, I am extremely pleased with the feedback that was given to me regarding my occupancy counter. I believe that the occupancy counter will benefit the Cal Poly community as a whole when looking for a place to gather, study, or catch up. I ran into a couple issues when implementing the Occupancy Counter. One of the main challenges I faced was not having access to the Occuspace devices so I had to create and work with mock data instead. This mock data is modeled after the Occuspace API so if I had access to the device, there wouldn't be that many changes (except calling to the API to gather the data). Furthermore, I ran into some initial issues with getting permission and access to the WordPress Admin site. Initially, I didn't know where to host my prototype, so it was hosted locally until I had access to WordPress. However, I did have issues with implementing my custom code, but I managed to get the right permissions and fix it.

If I had more time on my Occupancy Counter, I would implement booking the study rooms to the existing one the library had in place. However, due to the renovation of the library, the way to book study rooms isn't available at the moment. I also would update the floorplan to match the new library's floor plan and if I had permission, use the Occuspace device. For how I planned it in my blueprint, there will be forty Occuspace devices that span across every floor. Because of the way they are spaced and due to the number of Occuspace devices, it will give the best reading on each coordinate point for each floor. This will make it possible for heatmaps compared to a general occupancy percentage.

Conclusion

The need for an effective and user-friendly occupancy counter for the Kennedy Library is motivated by the challenges students face in finding available study spaces, especially during peak times like midterms and finals. The current solutions on campus are inconsistent and not easily accessible, leading to wasted time and frustration for students. This project aims to fill that gap by providing a more comprehensive and accessible solution that integrates real-time occupancy data directly on the Kennedy Library website.

The main shortcomings of the existing solutions, the inaccurate temporary study space counters and no current occupancy counter for the old and new library, are addressed by offering a detailed view of floor occupancy and study room availability. This allows students to plan their visits more effectively, reducing time spent searching for a spot and improving their overall study experience.

My work specifically focuses on creating a web-based occupancy counter that displays real-time data for each floor and study room in the remodeled Kennedy Library. By utilizing the Occuspace API and designing an intuitive interface, the application accomplishes its goal of making the library more accessible to students.

Moving forward, further validation through user testing and real-time data integration will help refine the application, ensuring it meets the needs of the student body and provides a valuable resource for navigating the Kennedy Library during busy times.

View the Occupancy Counter

View the User Testing Form

Occupancy Counter Code (Github)

Works Cited

Occuspace, https://web.occuspace.io/. Accessed 9 July 2024.

"Figma." *Figma*, 2024, https://www.figma.com/about/. Accessed 9 July 2024.

"Library Occupancy." *Cal State LA*, 2024, https://www.calstatela.edu/library/library-occupancy. Accessed 10 August 2024.

"Occuspace API." *Occuspace*, 2024, https://occuspace.io/api. Accessed 9 July 2024.